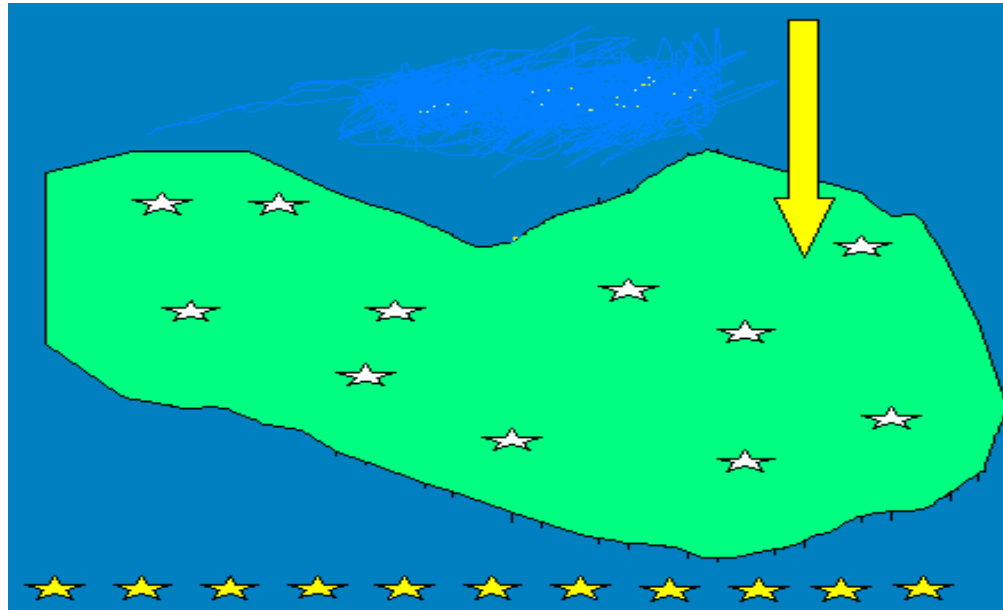


Hypothesis Test

“ **Hypothesis Test**”: A procedure for deciding between two hypotheses (null hypothesis – alternative hypothesis) on the basis of observations in a random sample

Sampling models

A. One sample model



- One sample model usually concerns with an intervention on population: If the intervention should make some change in population?

One – sample Hypothesis test

- **Compare proportion to a given value of rate**
- **Compare mean value to a given value of expectation**

- **Example 1.** If in Ha Dong District less than 90% motorcyclists use helms?
- **Example 2.** If proportion of girl-students equals to 50% ?
- **Example 3.** If in Viet Nam bred feeding is popular among more than 70% women?

Test 1. *Compare proportion to a given rate*

Two tails (sides) test:

(Null) Hypothesis

$$H_0: p = q$$

Alternative Hypothesis

$$H_a: p \text{ differs from } q$$

Lower (left) tail test:

(Null) Hypothesis

$$H_0: p \geq q$$

Alternative Hypothesis

$$H_a: p < q$$

Upper (right) tail test:

(Null) Hypothesis

$$H_0: p \leq q$$

Alternative Hypothesis

$$H_a: p > q$$

Test 1. *Compare proportion to a given rate*

$(X_1, X_2, \dots, X_n)$ - a sample of n independent observations collected from a binary variable X taking value 1 with (unknown) probability p ($0 < p < 1$) and value 0 with probability $1 - p \rightarrow$ Given a number q , how to have a conclusion comparing p with q based on information of the sample?

Solution

By Moivre-Laplace Theorem, for large sample size,

$$n \times p \geq 5 \text{ and } n \times (1-p) \geq 5,$$

sample proportion $m(p)/n$ of appearance of number 1 has distribution approximate to normal distribution with expectation p and variance $p \times (1-p)/n$. Then a testing procedure can be as follows:

Step 1. Estimate a sample proportion by

$$\hat{p} = m(p) / n$$

Calculate **z-test statistic**

$$z_q = (\hat{p} - q) / [\hat{p} \times (1 - \hat{p}) / n]^{1/2}$$

Two tails (sides) test:

(Null) Hypothesis

$$\mathbf{H}_0: p = q$$

Alternative Hypothesis

$$\mathbf{H}_a: p \neq q$$

P-value Approach

Step 2. (two sides test)

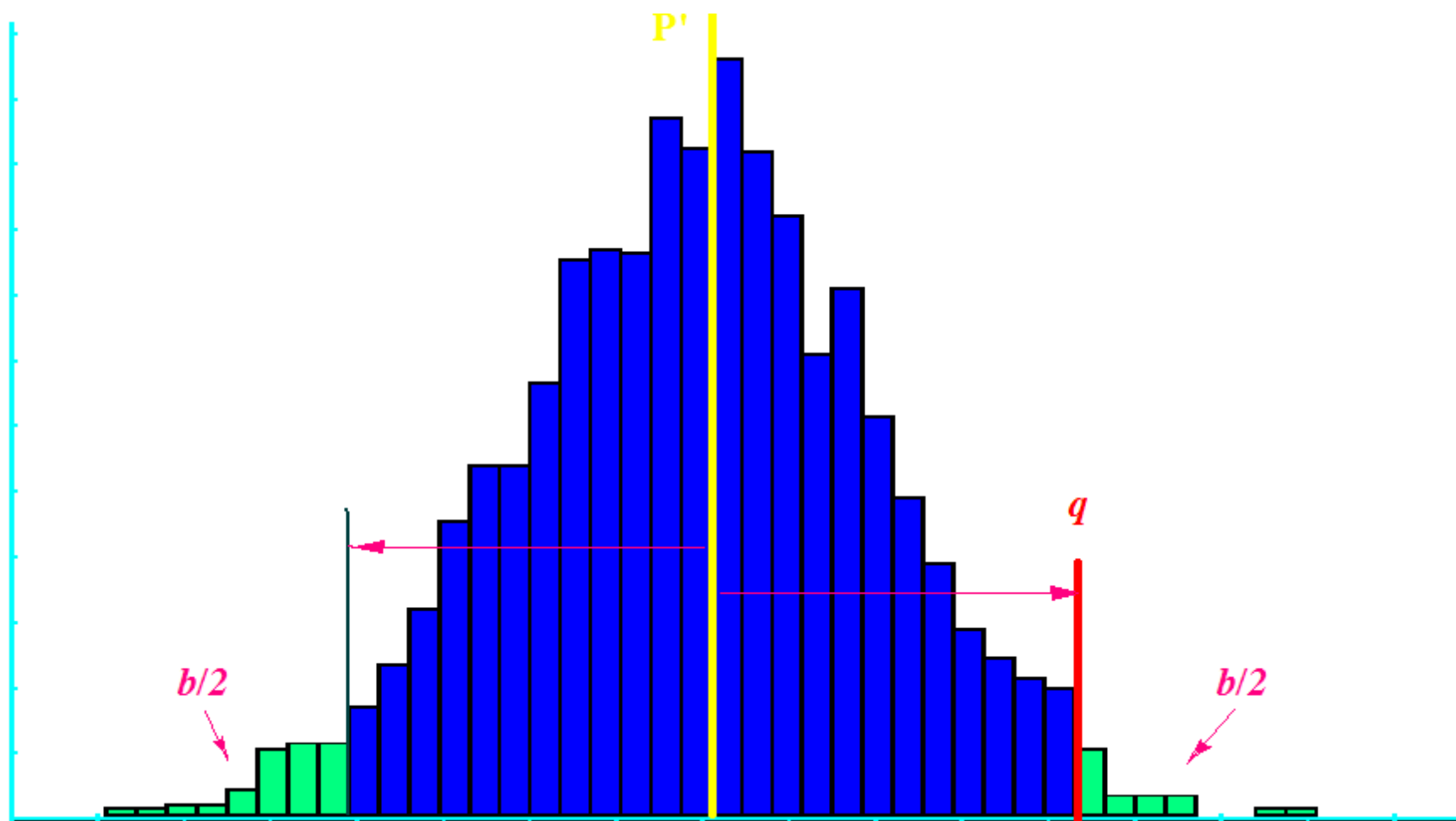
Approach A (p-value method): Calculate the probability such that the distance from estimate point to $\hat{p}$ should be more than $|q - \hat{p}|$:

$$b = P\{|Y - q| > |\hat{p} - q|\} = P\{|Z| > |z_q|\}$$

where Y is a normal variable with mean $\hat{p}$ and variance $\hat{p} \times (1 - \hat{p})/n$,

while Z is the standard normal variable (the normal variable with mean 0 and variance 1).

b = Probability (*p-value*) of wrong decision of excluding estimation value q (saying that q differs from true value of p) when this value should be a “good” value of estimation

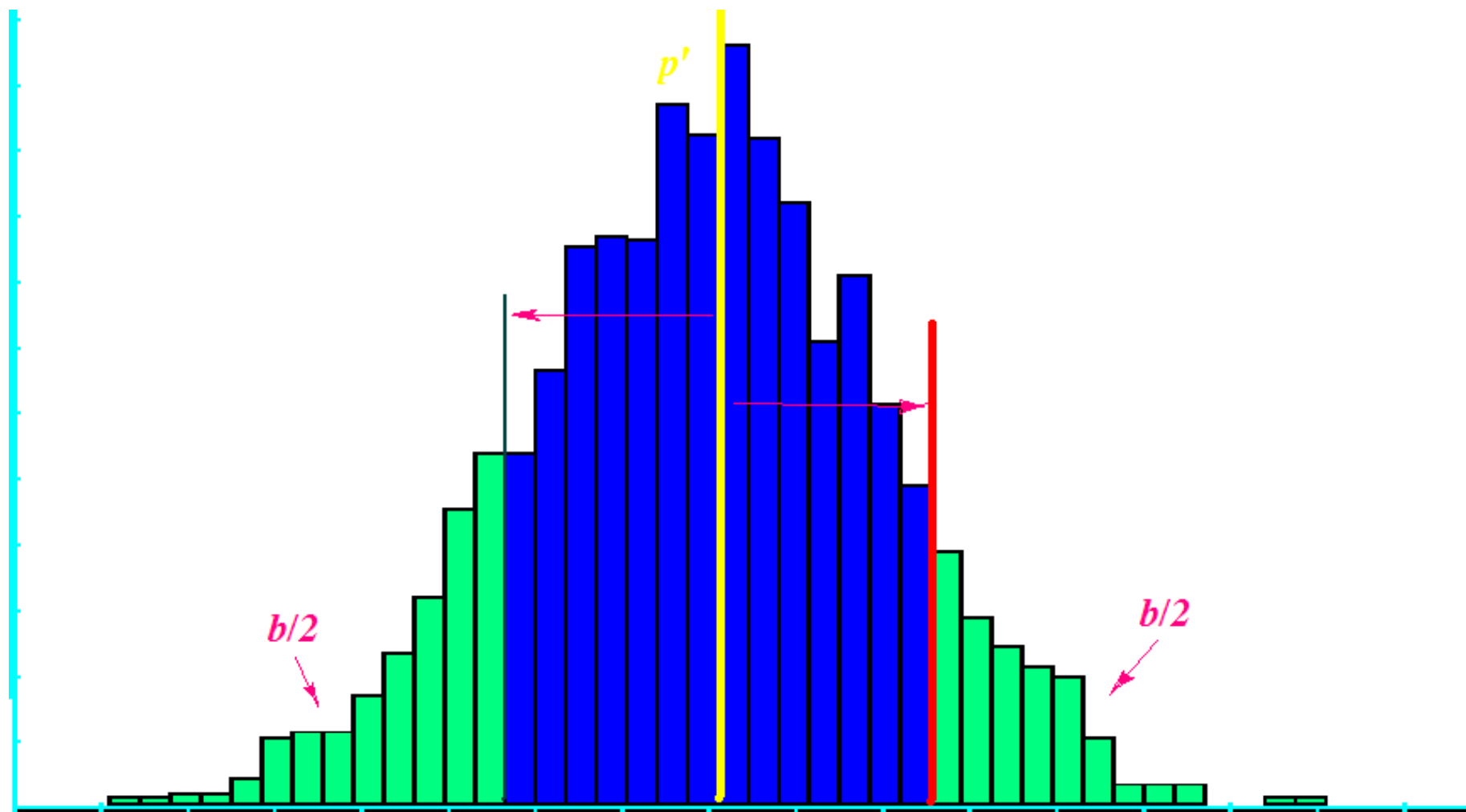


Step 3. (two sides test)

Compare b to the significance level α
($\alpha = 10\%, 5\%, 1\%, \dots$)

- If $b \leq \alpha \rightarrow$ ***reject the hypothesis*** H_0 , conclude that q differs from p , because possibility of getting mistake in decision is “very small”

- * If $b > \alpha \rightarrow$ ***accept the hypothesis*** H_0 , confirm $q = p$, because possibility of having mistake by rejecting the hypothesis is too high.



Two tails (sides) test:

(Null) Hypothesis

$$H_0: p = q$$

Alternative Hypothesis

$$H_a: p \neq q$$

Critical value Approach

Step 2. (two sides test)

Approach B (critical value method): With the given significance level α ($\alpha = 10\%, 5\%, 1\%, \dots$), calculate the critical value $z_{1-\alpha/2}$, the number point such that $1-\alpha/2$ is the probability of the standard normal variable Z taking values smaller or equal to $z_{1-\alpha/2}$:

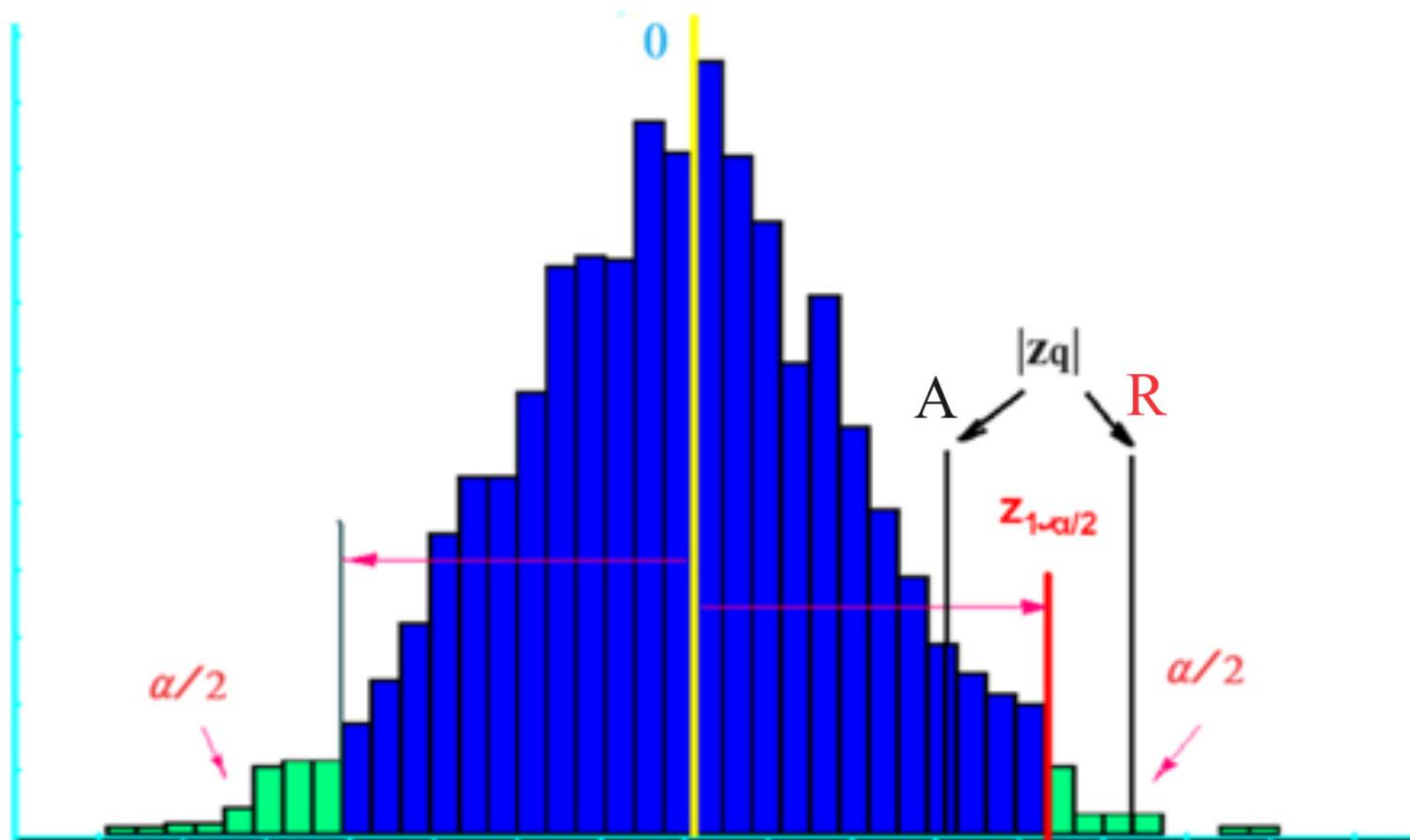
$$1-\alpha/2 = P\{Z \leq z_{1-\alpha/2}\}$$

Step 3. (two sides test)

Compare **absolute value** of the **z-test statistic** ($|z_q|$)
too the **critical value** $z_{1-\alpha/2}$
($\alpha = 10\%, 5\%, 1\%, \dots$)

- If $|z_q| \geq z_{1-\alpha/2} \rightarrow$ **reject the hypothesis** H_0 ,
conclude that **q** differs from **p** .

- * If $|z_q| < z_{1-\alpha/2} \rightarrow$ **accept the hypothesis** H_0 ,
confirm **$q = p$** .



Two tails (sides) test:

(Null) Hypothesis

$$\mathbf{H_0: } p = q$$

Alternative Hypothesis

$$\mathbf{H_a: } p \neq q$$

Confidence Interval Approach

Approach C. Using confidence interval

With **confidence level** of **95%**, we can use confidence interval for hypothesis testing:

$$\left[\hat{p} - 1.96 * \underbrace{\sqrt{\hat{p} \cdot (1 - \hat{p}) / n}}_{\text{standard error}}; \hat{p} + 1.96 * \underbrace{\sqrt{\hat{p} \cdot (1 - \hat{p}) / n}}_{\text{margin of error}} \right]$$



Decide

- **Reject the Hypothesis H** if the confidence interval does not contain the point **q**
- **Accept the Hypothesis H** if the confidence interval contains the point **q**

Using Excel to Compute Normal P-value and Critical Value

- Excel has two functions for computing P-value and critical respect to normal distribution:
 - NORMDIST, NORMSDIST are used to compute the P-values.
 - NORMINV and NORMSINV are used to compute the Critical values.

**Compare mean value to a given
value of expectation**

Test 2. Compare mean value to a given value of expectation

Problem: Taking a sample from a variable X with normal distribution, we need to compare the expectation μ of X to a given value u .

Hypothesis

$$H_0: \mu = u$$

Alternative Hypothesis

$$H_a: \mu \text{ differs } (\neq, \leq, \geq) \text{ from } u$$

Two tails (sides) test:

(Null) Hypothesis

$$H_0: \mu = u$$

Alternative Hypothesis

$$H_a: \mu \text{ differs from } u$$

Lower (left) tail test:

(Null) Hypothesis

$$H_0: \mu \geq u$$

Alternative Hypothesis

$$H_a: \mu < u$$

Upper (right) tail test:

(Null) Hypothesis

$$H_0: \mu \leq u$$

Alternative Hypothesis

$$H_a: \mu > u$$

**Compare mean value when
variance of variable is unknown**

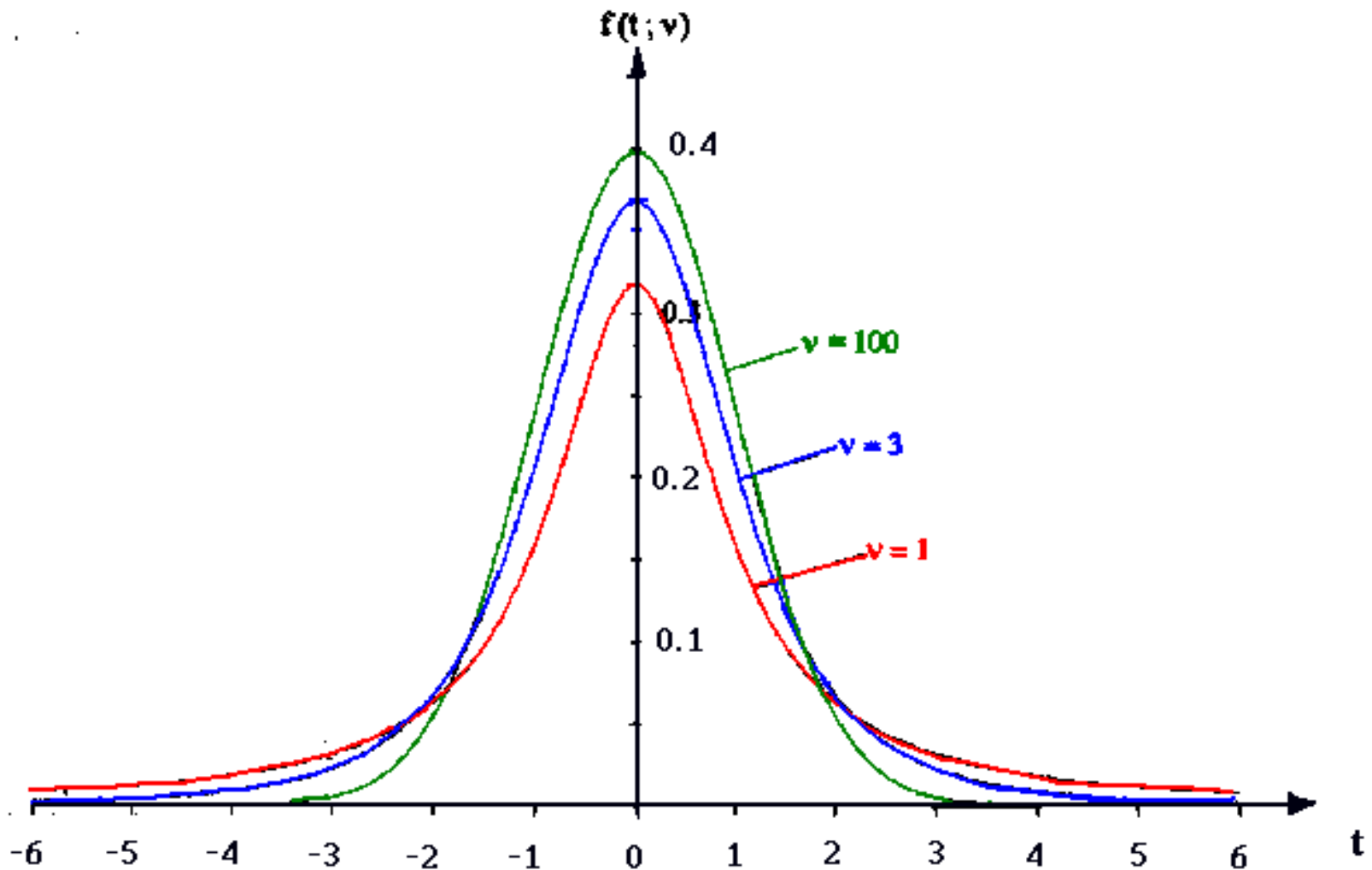
For testing the above hypothesis, the distribution of sample mean value must be known. When **variance of variable is unknown** the following theorem can be applied:

Theorem. Let $(X_1, X_2, \dots, X_n)$ be a sample of n independent observations taken from a normal distributed variable X with expectation μ , $\bar{X}$ is sample mean value and S^2 is sample variance. Then the (new) variable

$$t = \frac{\sqrt{n-1}}{S} \cdot (\bar{X} - \mu)$$

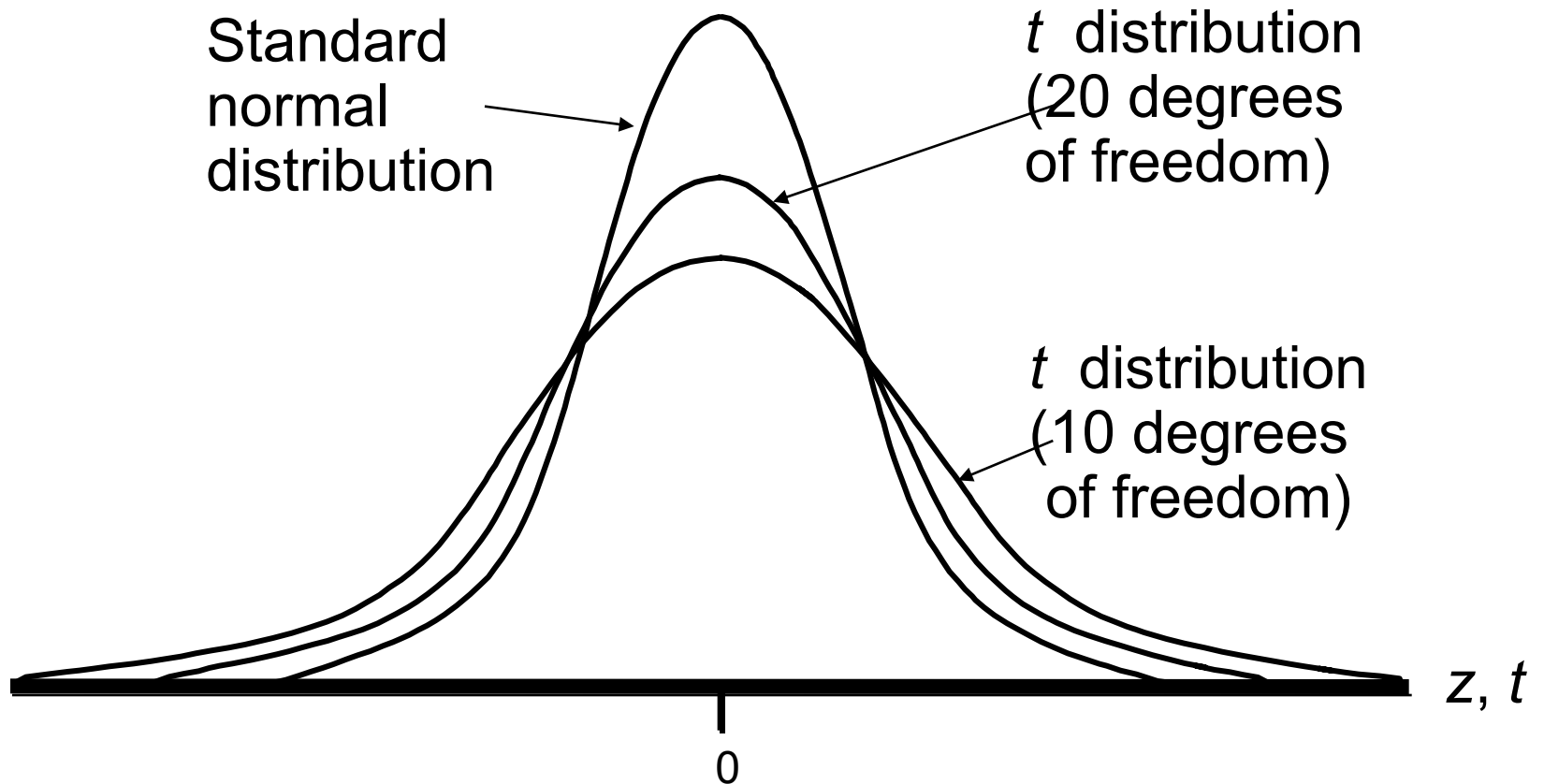
has T-Student distribution with $(n-1)$ degrees of freedom.

Student (T) distribution



Parameter of Student Distribution: “Degree of Freedom” v

t Distribution



t Distribution

- For more than 100 degrees of freedom, the standard normal z value provides a good approximation to the t value.

Steps of Testing

Step 1. Estimate sample Mean Value $\text{Mean}(X)$ Standard Deviation $\text{SD}(X)$

Step 2. Calculate the statistic

$$t(u) = \frac{\sqrt{n-1} \cdot (\text{Mean}(X) - u)}{\text{SD}(X)}$$

where n is sample size and u is the given value to which the mean value has be compared

Two tails (sides) test:

(Null) Hypothesis

$$H_0: \mu = u$$

Alternative Hypothesis

$$H_a: \mu \text{ differs from } u$$

P-value Approach

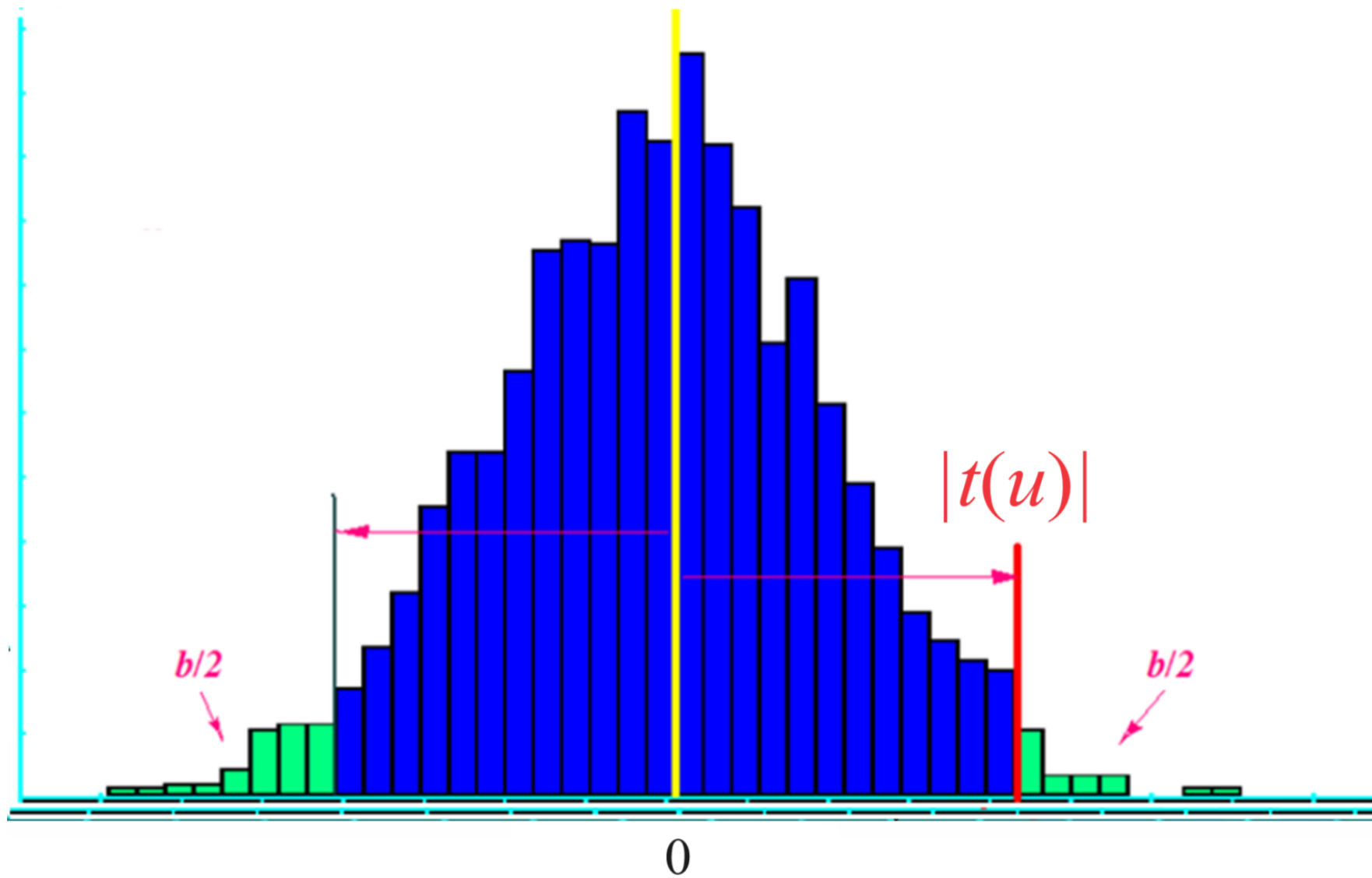
Step 2. (two sides test)

Approach A (p-value method): Calculate the probability

$$b = P\{|T(n-1)| \geq |t(u)|\},$$

where $T(n-1)$ is a variable which has Student distribution with $(n-1)$ degrees of freedom.

b = Probability (*p-value*) of wrong decision of excluding estimation value u when this value should be a “good” value of estimation

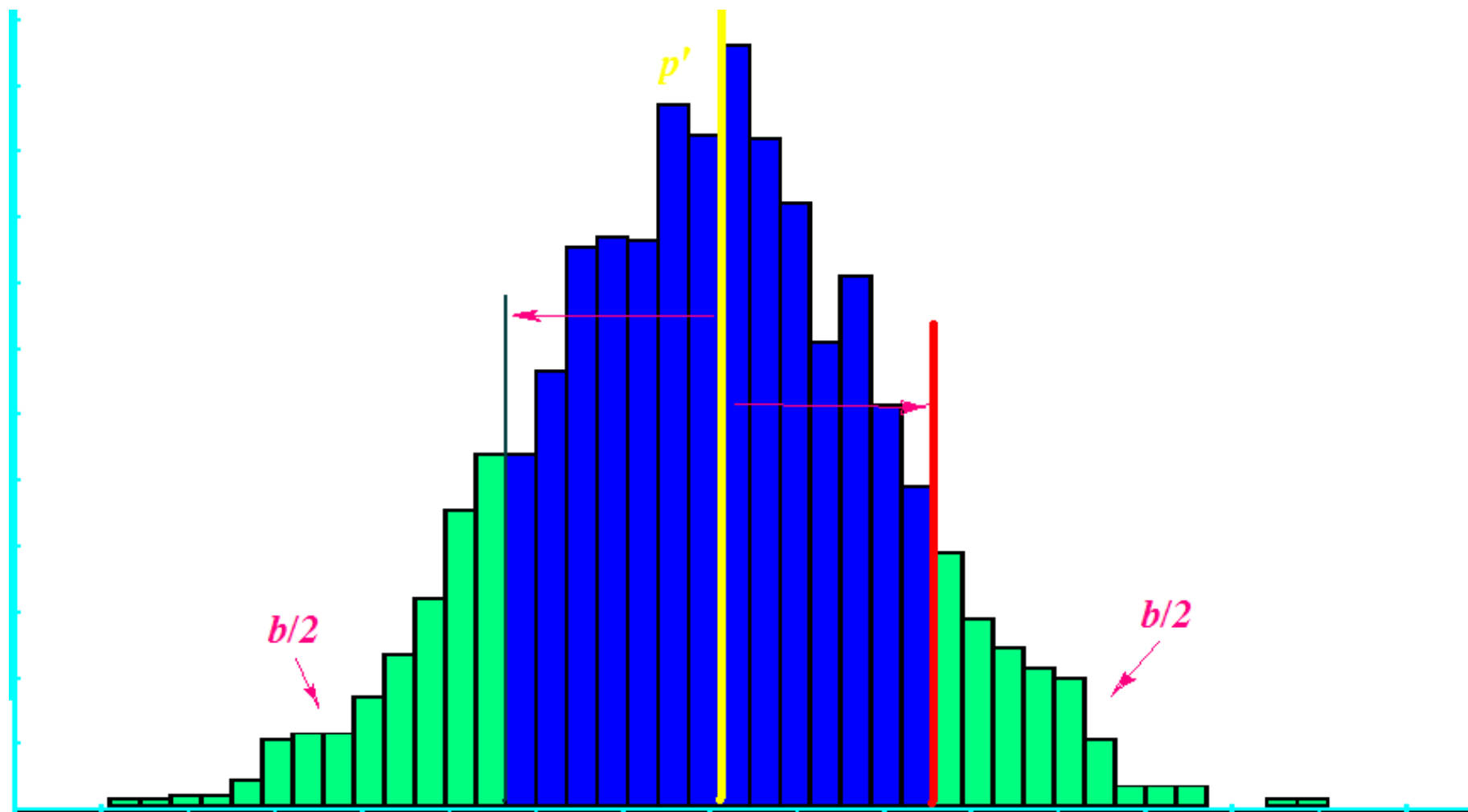


Step 3. (two sides test)

Compare b to the significance level α
($\alpha = 10\%, 5\%, 1\%, \dots$)

- If $b \leq \alpha \rightarrow$ ***reject the hypothesis*** H_0 , conclude that μ differs from u , because possibility of getting mistake in decision is “very small”

- * If $b > \alpha \rightarrow$ ***accept the hypothesis*** H_0 , confirm $\mu = u$, because possibility of having mistake by rejecting the hypothesis is too high.



Two tails (sides) test:

(Null) Hypothesis

$$H_0: \mu = u$$

Alternative Hypothesis

$$H_a: \mu \text{ differs from } u$$

Critical value Approach

Step 2. (two sides test)

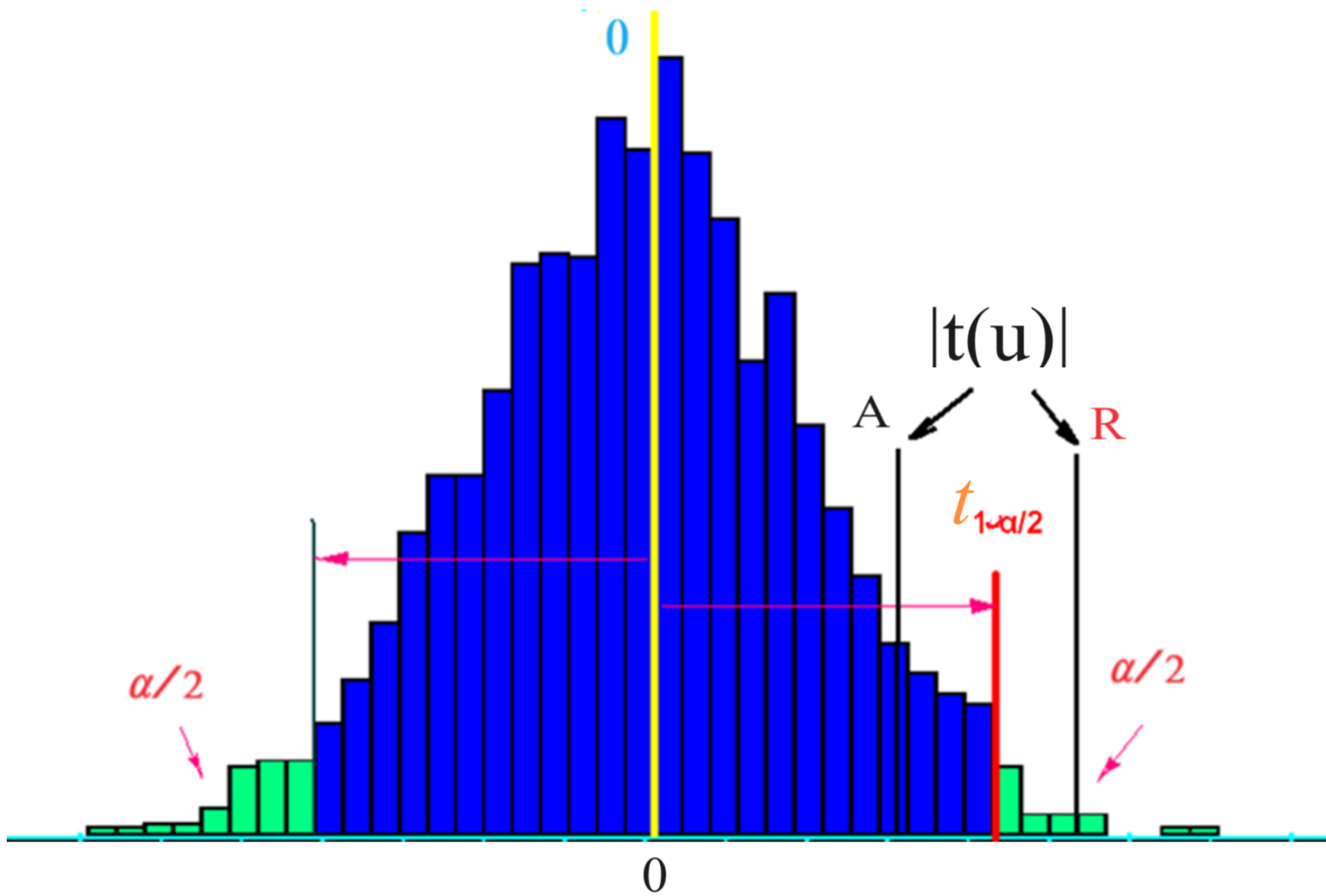
Approach B (critical value method): With the given significance level α ($\alpha = 10\%, 5\%, 1\%, \dots$), calculate the critical value $t_{1-\alpha/2}$, the number point such that $1-\alpha/2$ is the probability of the Student distributed variable $T(n-1)$ taking values smaller or equal to $t_{1-\alpha/2}$.

$$1-\alpha/2 = P\{T(n-1) \leq t_{1-\alpha/2}\}$$

Step 3. (two sides test)

Compare **absolute value** of the **t-test statistic** ($|t(u)|$)
too the **critical value** $t_{1-\alpha/2}$
($\alpha = 10\%, 5\%, 1\%, \dots$)

- If $|t(u)| \geq t_{1-\alpha/2} \rightarrow$ ***reject the hypothesis*** H_0 ,
conclude that μ differs from u .
- * If $|t(u)| < t_{1-\alpha/2} \rightarrow$ ***accept the hypothesis*** H_0 ,
confirm $\mu = u$.



Two tails (sides) test:

(Null) Hypothesis

$$H_0: \mu = u$$

Alternative Hypothesis

$$H_a: \mu \text{ differs from } u$$

Confidence Interval Approach

Approach C. Using confidence interval

With **confidence level** of $(1-\alpha)$, we can use confidence interval for hypothesis testing:

$$\left[\underbrace{\text{Mean}(X) - T(n-1, 1-\alpha/2) * \text{SD}(X) / \sqrt{n-1}}_{\text{margin of error}} ; \underbrace{\text{Mean}(X) + T(n-1, 1-\alpha/2) * \text{SD}(X) / \sqrt{n-1}}_{\text{standard error}} \right]$$



Decide

- **Reject the Hypothesis H** if the confidence interval does not contain the point **u**
- **Accept the Hypothesis H** if the confidence interval contains the point **u**

Using Excel to Compute Student P-value and Critical Value

- Excel has two functions for computing P-value and critical respect to Student distribution:
 - TDIST is used to compute the P-values.
 - TINV is used to compute the Critical values.